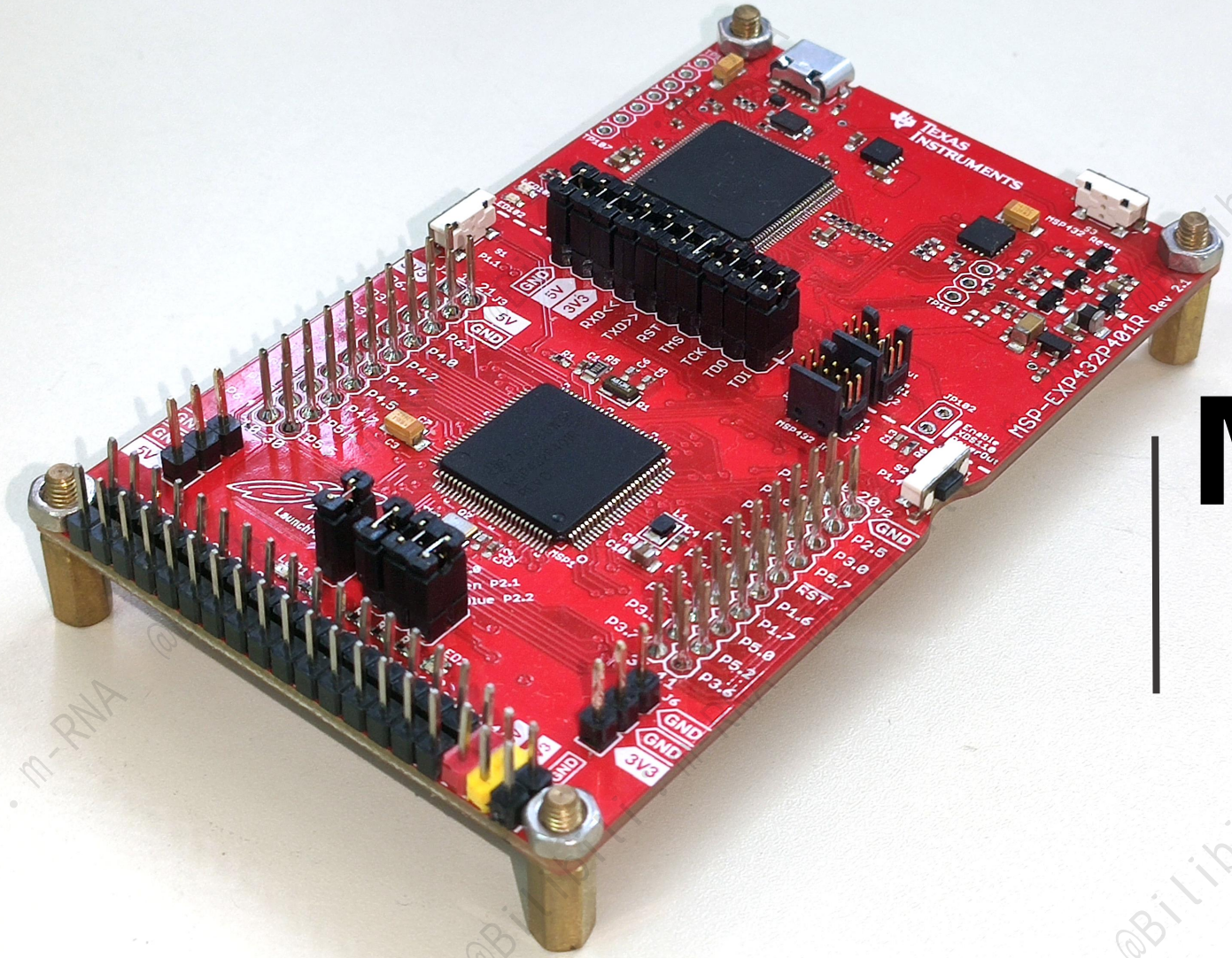


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第0讲 序章

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适用对象

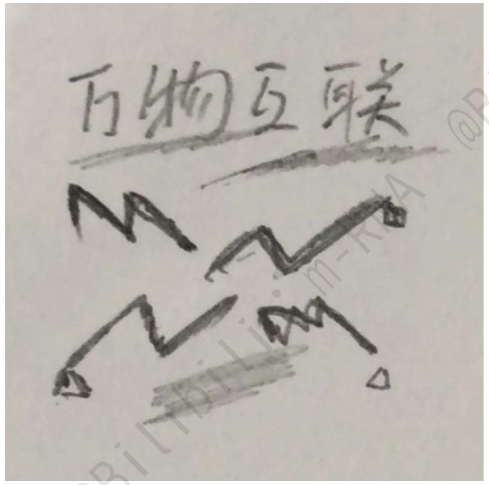
有基础	学习中	没基础
光速速成	对比学习	不懂会用

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UP



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目录

第0讲 序章
第1讲 配置开发环境
第2讲 GPIO
第3讲 时钟简介
第4讲 滴答计时器延时
第5讲 外部中断
第6讲 NVIC中断优先级管理
第7讲 串口实验
第8讲 定时器A
第9讲 定时器32
第10讲 OLED显示(IIC)
第11讲 ADC模数转换
第12讲 DMA传输

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第1讲 配置开发环境

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01

安装Keil、Pack包

解压安装
TexasInstruments.MSP432P4xx_DFP.3.2.6

02

安装CSS以获得驱动

解压安装
CCS10.4.0.00006_win64

03

获取SDK软件开发套件

解压安装
simplelink_msp432p4_sdk_3_40_01_02

扫码下载文件

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01

评估板手册

slau597f.pdf
有评估板原理图

02

芯片数据手册

msp432p401r.pdf
引脚功能

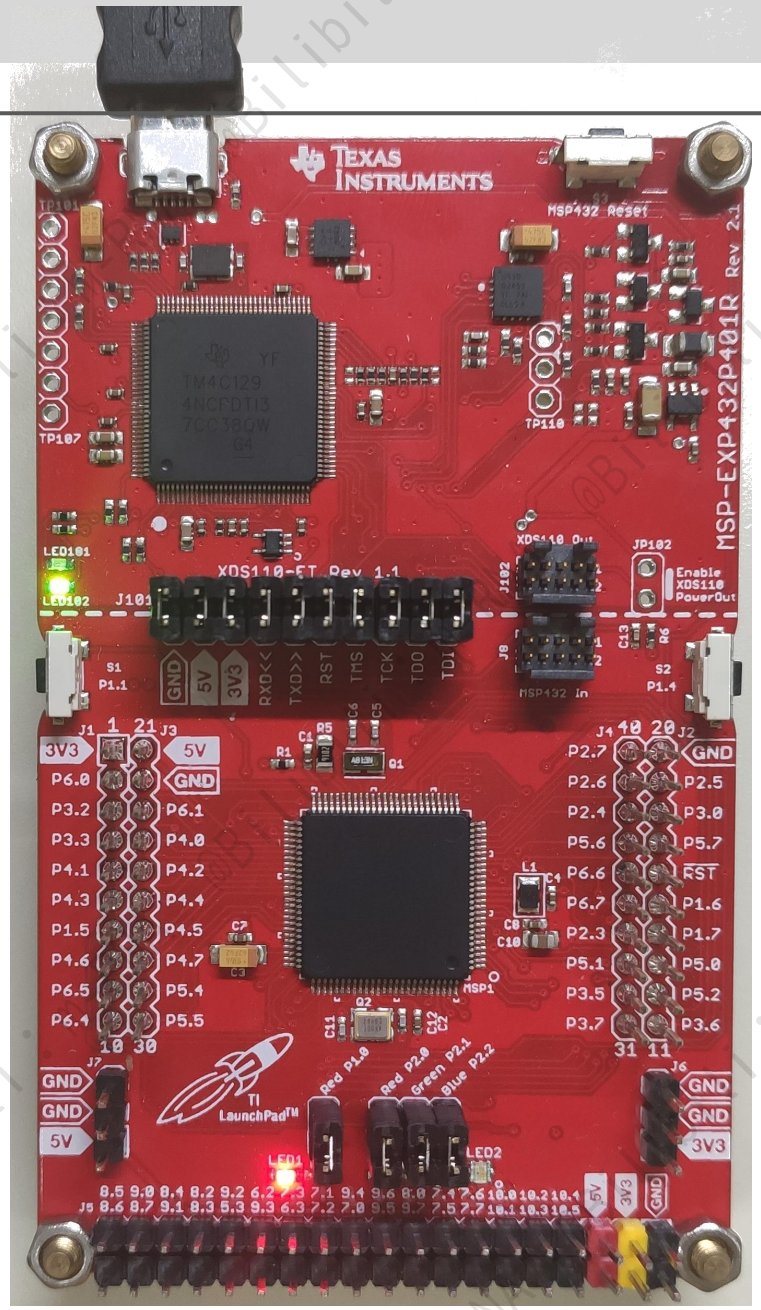
03

芯片技术手册

slau356h.pdf
编程配置参考

扫码下载文件

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TEXAS INSTRUMENTS
MSP-EXP432P401R Rev 2.1
MSP432 Reset
XDS110-ET Rev 1.1
Enable XDS110 PowerOut
MSP432 in
TM4C129
LaunchPad™
Red P1.0
Red P2.0
Green P2.1
Blue P2.2

2-1 GPIO输出

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参考资料

msp432p402.pdf P17

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配置GPIO模式

GPIO_setAsOutputPin(Port, Pin);

设置高低电平

GPIO_setOutputHighOnPin(Port, Pin);
GPIO_setOutputLowOnPin(Port, Pin);
GPIO_toggleOutputOnPin(Port, Pin);

配置驱动强度

GPIO_setDriveStrengthHigh(Port, Pin);
GPIO_setDriveStrengthLow(Port, Pin);

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2-2 GPIO输入

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配置GPIO模式

GPIO_setAsInputPin(Port, Pin);
GPIO_setAsInputPinWithPullUpResistor(Port, Pin);
GPIO_setAsInputPinWithPullDownResistor(Port, Pin);

获取电平状态

GPIO_getInputPinValue(Port, Pin);

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2-3 GPIO复用

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配置GPIO模式

GPIO_setAsPeripheralModuleFunctionInputPin(Port, Pin,mode);
GPIO_setAsPeripheralModuleFunctionOutputPin(Port, Pin,mode);

mode参数有效值

GPIO_PRIMARY_MODULE_FUNCTION
GPIO_SECONDARY_MODULE_FUNCTION
GPIO_TERTIARY_MODULE_FUNCTION

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3-1 时钟简介

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3-2 配置外部晶振

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配置时钟频率

CS_setExternalClockSourceFrequency(lfxtFreq, hfxtFreq);

配置驱动模式

CS_startHFXT(bypassMode);
CS_startLFXT(bypassMode);

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配置内核电压等级

PCM_setCoreVoltageLevel(voltageLevel);

设置Flash等待状态

FlashCtl_setWaitState(bank, waitState);

初始化时钟

CS_initClockSignal(clk,clkSource,clkSourceDiv)

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获取时钟频率

CS_getACLK();
CS_getMCLK();
CS_getSMCLK();
CS_getHSMCLK();

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第4讲 滴答计时器延时

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第5讲 外部中断

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参考资料

msp432p401r.pdf　P17

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gpio.h

1 开启外部中断

GPIO_enableInterrupt(GPIO_PORT_Px, GPIO_PINx);

2 配置触发方式

GPIO_interruptEdgeSelect(GPIO_PORT_P1, GPIO_PIN4, **Edge**);

Edge有效值

GPIO_HIGH_TO_LOW_TRANSITION
GPIO_LOW_TO_HIGH_TRANSITION

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gpio.h

3 获取GPIO中断状态

GPIO_getEnabledInterruptStatus(GPIO_PORT_Px);

4 清除GPIO中断标志位

GPIO_clearInterruptFlag(GPIO_PORT_Px, GPIO_PINx);

配合使用

status = GPIO_getEnabledInterruptStatus(GPIO_PORT_Px);

GPIO_clearInterruptFlag(GPIO_PORT_Px, **status**);

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interrupt.h

5 开启总中断

Interrupt_enableMaster(void);

6 开启端口中断

Interrupt_enableInterrupt(interruptNumber);

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一般配置步骤

1.配置GPIO输入
2.清除中断标志位
3.配置触发方式
4.开启外部中断
5.开启端口中断
6.开启总中断
7.编写GPIO ISR

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6-1 中断优先级管理

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参考资料

slau356h.pdf　　P82

msp432p401r.pdf　P117

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例子

系统有两个中断，中断A和中断B，中断号分别为1和2。

当不进行中断优先级配置时，组优先级一致，中断号即为中断优先级，中断号小的中断优先级高，所以中断优先级为A>B。假如此时系统正在执行中断B，而中断A发生了，系统会如何处理呢？因为它们组优先级一样，故中断A不能打断中断B，系统会先挂起中断A，待中断B执行完后，再执行中断A；

倘若将中断A的组优先级设置为1，中断B的组优先级设置为2，此时系统正在执行中断B，而中断A发生了，系统会如何处理呢？因为组优先级小的优先级高，所以中断优先级是A>B，故系统打断中断B，执行中断A，待中断A执行完后，再继续执行中断B。

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interrupt.h

设置组优先级

Interrupt_setPriority(interruptNumber, **level**);

level： x<<5 , x∈(0,7)
只使用高3位,配置时得左移5位

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6-2 软件挂起中断
6-3 屏蔽中断优先级

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interrupt.h

1 软件挂起中断

Interrupt_pendInterrupt(interruptNumber);

2 屏蔽中断优先级

Interrupt_setPriorityMask(level);

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第7讲 串口配置

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参考资料

slau597f.pdf P38

msp432p401r.pdf P6

slau356h.pdf P7 & P904

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3 Device Comparison

Table 3-1 summarizes the features of the MSP432P401x microcontrollers.

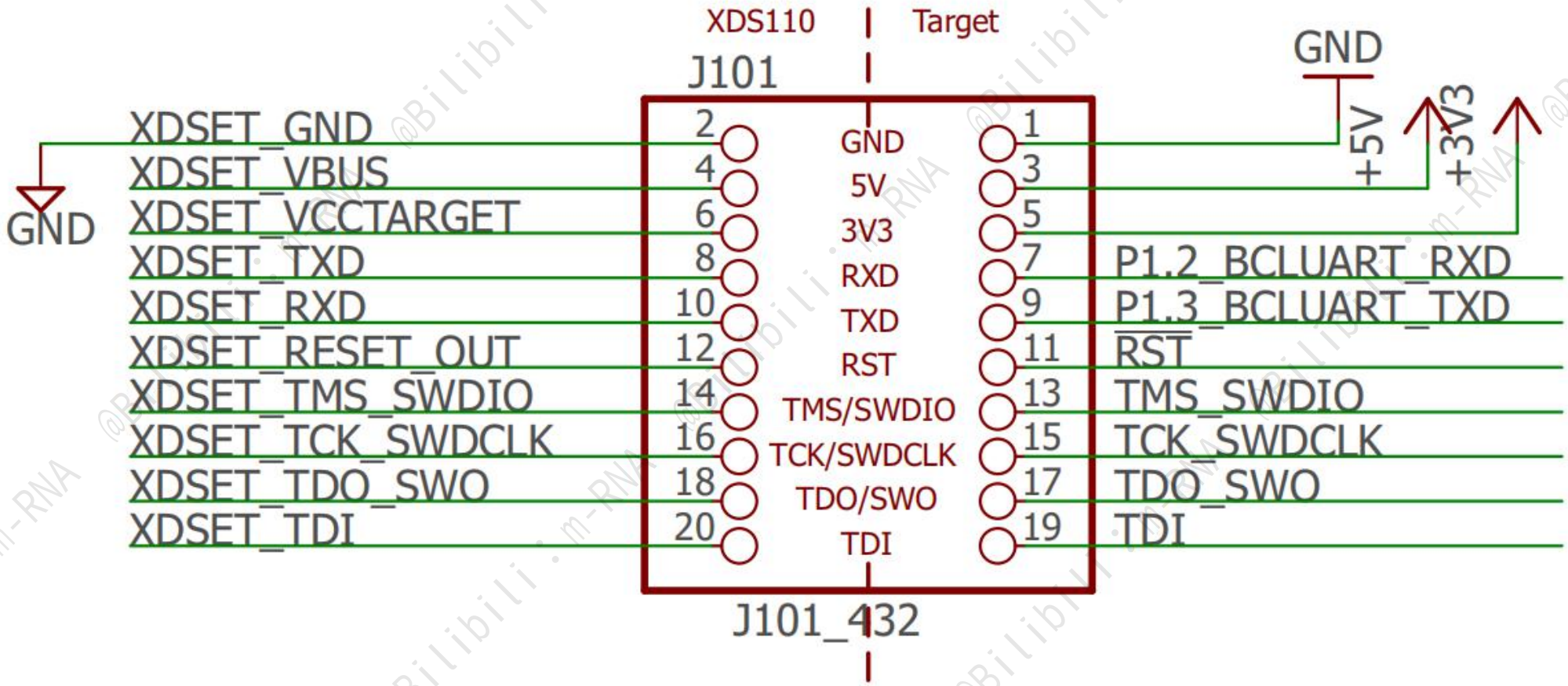
Table 3-1. Device Comparison[1]

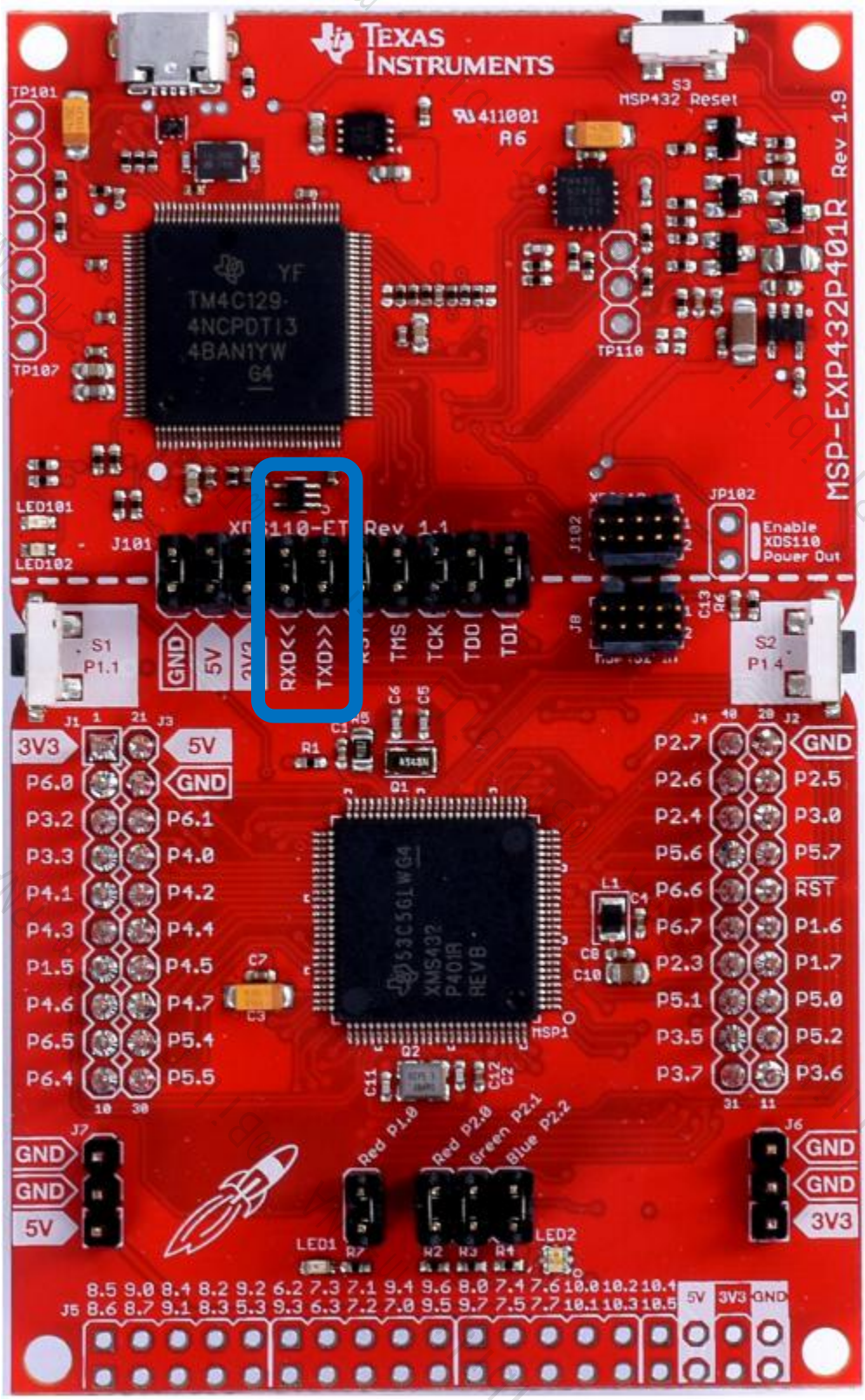
DEVICE	FLASH (KB)	SRAM (KB)	Precision ADC (Channels)	COMP_E0 (Channels)	COMP_E1 (Channels)	Timer_A[2]	eUSCI_A: UART, IrDA, SPI	eUSCI_B: SPI, I2C	20-mA DRIVE I/O	TOTAL I/Os	PACKAGE
MSP432P401RIPZ	256	64	24 ext, 2 int	8	8	5, 5, 5, 5	4	4	4	84	100 PZ
MSP432P401MIPZ	128	32	24 ext, 2 int	8	8	5, 5, 5, 5	4	4	4	84	100 PZ
MSP432P401RIZXH	256	64	16 ext, 2 int	6	8	5, 5, 5	3	4	4	64	80 ZXH
MSP432P401MIZXH	128	32	16 ext, 2 int	6	8	5, 5, 5	3	4	4	64	80 ZXH
MSP432P401RIRGC	256	64	12 ext, 2 int	2	4	5, 5, 5	3	3	4	48	64 RGC
MSP432P401MIRGC	128	32	12 ext, 2 int	2	4	5, 5, 5	3	3	4	48	64 RGC

MSP432P401R串口资源		
串口	RXD	TXD
EUSCI_A0_BASE	P1.2	P1.3
EUSCI_A1_BASE	P2.2	P2.3
EUSCI_A2_BASE	P3.2	P3.3
EUSCI_A3_BASE	P9.6	P9.7

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XDS110-ET <<---->> LaunchPad




TEXAS INSTRUMENTS
MSP-EXP432P401R Rev 1.9
TM4C129-
4NCPDTI3
4BAN1YW
G4
XDS110-ET Rev 1.1
RXD<<
TXD>>
GND
5V
TMS
TCK
TDO
TDI
3V3
P6.0
P3.2
P3.3
P4.1
P4.3
P1.5
P4.6
P6.5
P6.4
P6.1
P4.0
P4.2
P4.4
P4.5
P4.7
P5.4
P5.5
P2.7
P2.6
P2.4
P5.6
P6.6
P6.7
P2.3
P5.1
P3.5
P3.7
P2.5
P3.0
P5.7
RST
P1.6
P1.7
P5.0
P5.2
P3.6
Red P1.0
Red P2.0
Green P2.1
Blue P2.2
LED1
LED2
Enable
XDS110
Power Out
MSP432 Reset

24.1 Enhanced Universal Serial Communication Interface A (eUSCI_A) Overview

The eUSCI_A module supports two serial communication modes:

- UART mode
- SPI mode

24.2 eUSCI_A Introduction – UART Mode

In asynchronous mode, the eUSCI_Ax modules connect the device to an external system through two external pins, UCAxRXD and UCAxTXD. UART mode is selected when the UCSYNC bit is cleared.

UART mode features include:

- 7-bit or 8-bit data with odd, even, or no parity
- Independent transmit and receive shift registers
- Separate transmit and receive buffer registers
- LSB-first or MSB-first data transmit and receive
- Built-in idle-line and address-bit communication protocols for multiprocessor systems
- Programmable baud rate with modulation for fractional baud-rate support
- Status flags for error detection and suppression
- Status flags for address detection
- Independent interrupt capability for receive, transmit, start bit received, and transmit complete

Figure 24-1 shows the eUSCI_Ax when configured for UART mode.

UART模式的特性包括:

◇7/8个数据位、1个奇/偶/无奇偶校验位;
◇独立的发送和接收移位寄存器;
◇独立的发送和接收缓冲寄存器;
◇LSB优先/MSB优先的数据发送和接收;
◇为多处理器系统内置空闲线和地址位通信协议;
◇支持分数波特率的可编程调制波特率;
◇用于错误检测和抑制的状态标志;
◇针对地址检测的状态标志;
◇针对接收、发送,起始位接收和发送完成的独立中断能力。

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uart.h

1 初始化串口模块

UART_initModule(EUSCI_Ax_BASE, &uartConfig);

2 使能串口模块

UART_enableModule(EUSCI_Ax_BASE);

3 开启串口相关中断

UART_enableInterrupt(EUSCI_Ax_BASE, EUSCI_x_INTERRUPT);

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uart.h

4 获取数据

UART_receiveData(EUSCI_Ax_BASE);

5 发送数据

UART_transmitData(EUSCI_Ax_BASE，Data_8bit）；

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interrupt.h

6 开启串口端口中断

Interrupt_enableInterrupt(INT_EUSCIAx);

7 开启总中断

Interrupt_enableMaster(void);

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一般配置步骤

0.配置时钟
1.配置GPIO复用
2.配置结构体
3.初始化串口
4.开启串口
5.开启串口相关中断
6.开启串口端口中断
7.开启总中断
8.编写UART ISR

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对接标准输入输出库后

0.配置时钟
1.配置GPIO复用
2.配置UART结构体
3.初始化串口

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正确值

Clock: Custom Hz 3000000 Hz
Baud rate: 9600 bps
calculate

The recommended parameters for DriverLib

clockPrescalar: 19
firstModReg: 8
secondModReg: 85
overSampling: 1

Clock: Custom Hz 3000000 Hz
Baud rate: 19200 bps
calculate

The recommended parameters for DriverLib

clockPrescalar: 9
firstModReg: 12
secondModReg: 34
overSampling: 1

Clock: Custom Hz 48000000 Hz
Baud rate: 460800 bps
calculate

The recommended parameters for DriverLib

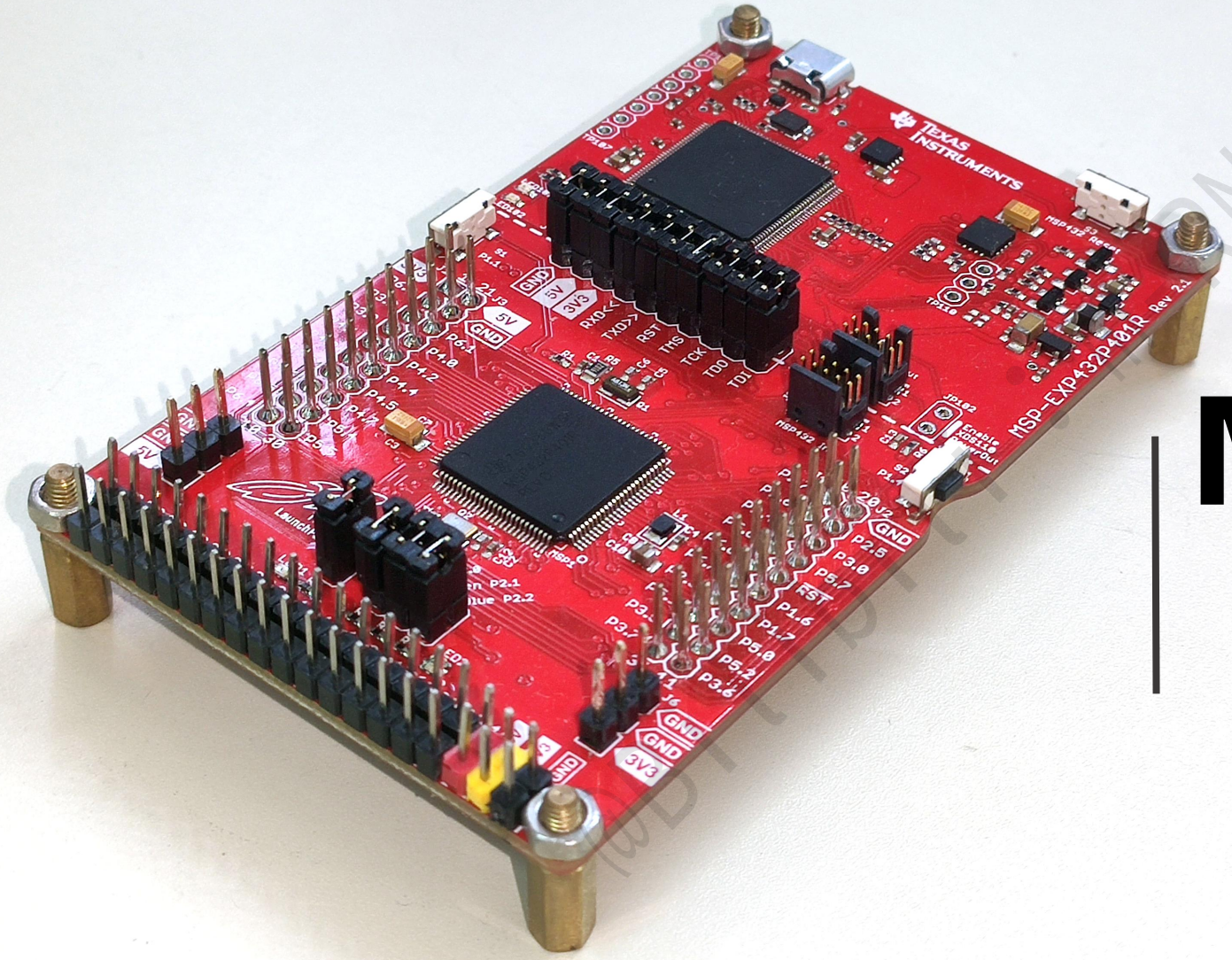
clockPrescalar: 6
firstModReg: 8
secondModReg: 32
overSampling: 1

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谢 谢

PPT Part1一开始是比较简陋的，大家见谅；
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2021/9

8-1 定时器A中断

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参考资料

slau356h.pdf P783

msp432p401r.pdf P6

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3 Device Comparison

Table 3-1 summarizes the features of the MSP432P401x microcontrollers.

Table 3-1. Device Comparison[1]

DEVICE	FLASH (KB)	SRAM (KB)	Precision ADC (Channels)	COMP_E0 (Channels)	COMP_E1 (Channels)	Timer_A[2]	eUSCI_A: UART, IrDA, SPI	eUSCI_B: SPI, I2C	20-mA DRIVE I/O	TOTAL I/Os	PACKAGE
MSP432P401RIPZ	256	64	24 ext, 2 int	8	8	5, 5, 5, 5	4	4	4	84	100 PZ
MSP432P401MIPZ	128	32	24 ext, 2 int	8	8	5, 5, 5, 5	4	4	4	84	100 PZ
MSP432P401RIZXH	256	64	16 ext, 2 int	6	8	5, 5, 5	3	4	4	64	80 ZXH
MSP432P401MIZXH	128	32	16 ext, 2 int	6	8	5, 5, 5	3	4	4	64	80 ZXH
MSP432P401RIRGC	256	64	12 ext, 2 int	2	4	5, 5, 5	3	3	4	48	64 RGC
MSP432P401MIRGC	128	32	12 ext, 2 int	2	4	5, 5, 5	3	3	4	48	64 RGC

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19.1 Timer_A Introduction

Timer_A is a 16-bit timer/counter with up to seven capture/compare registers. Timer_A can support multiple capture/compares, PWM outputs, and interval timing. Timer_A also has extensive interrupt capabilities. Interrupts may be generated from the counter on overflow conditions and from each of the capture/compare registers.

Timer_A features include:

- Asynchronous 16-bit timer/counter with four operating modes
- Selectable and configurable clock source
- Up to seven configurable capture/compare registers
- Configurable outputs with pulse width modulation (PWM) capability
- Asynchronous input and output latching

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Timer_A的特性包括:

◇具有4种操作模式的异步16位定时/计数器;
◇可选择和可配置的时钟源;
◇最多达7个可配置的捕获/比较模块;
◇具有PWM 功能的可配置输出;
◇异步输入和输出锁存。

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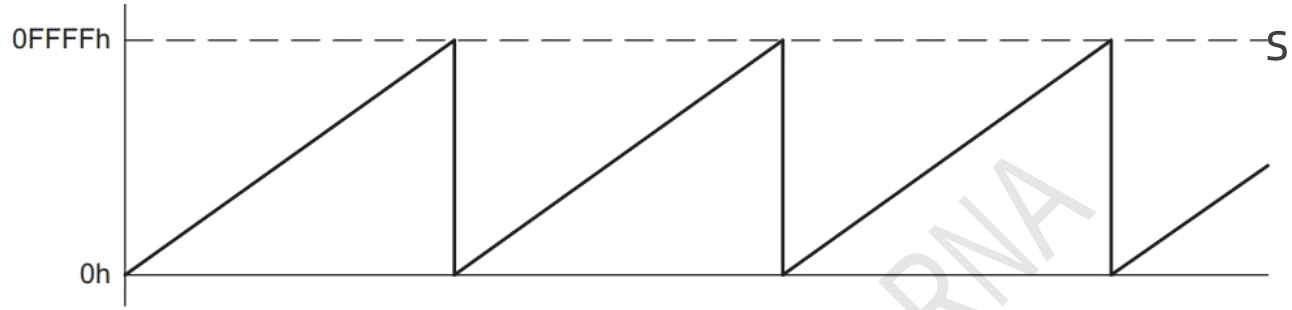


Figure 19-4. Continuous Mode

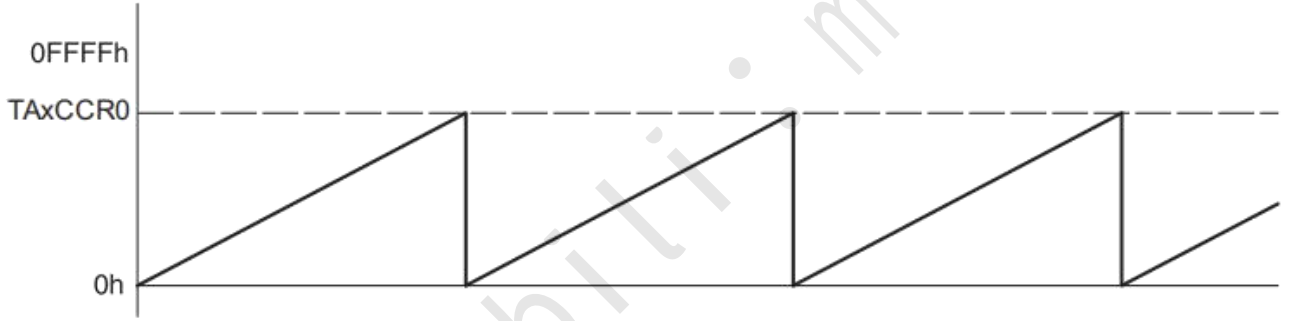


Figure 19-2. Up Mode

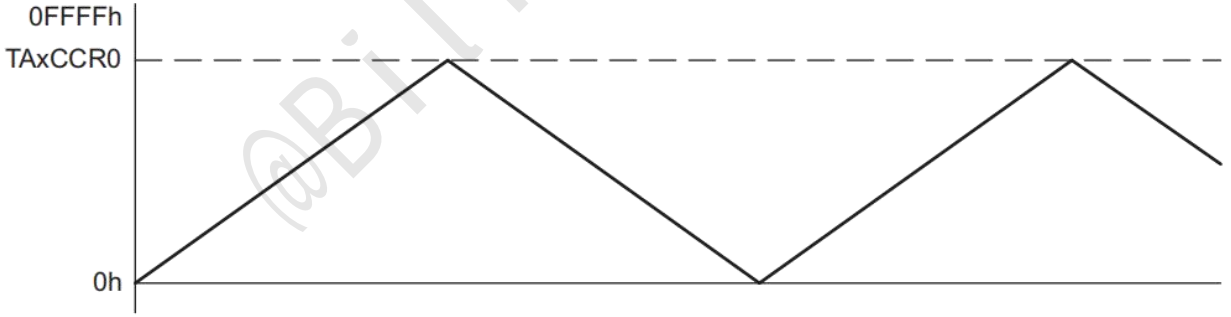


Figure 19-7. Up/Down Mode

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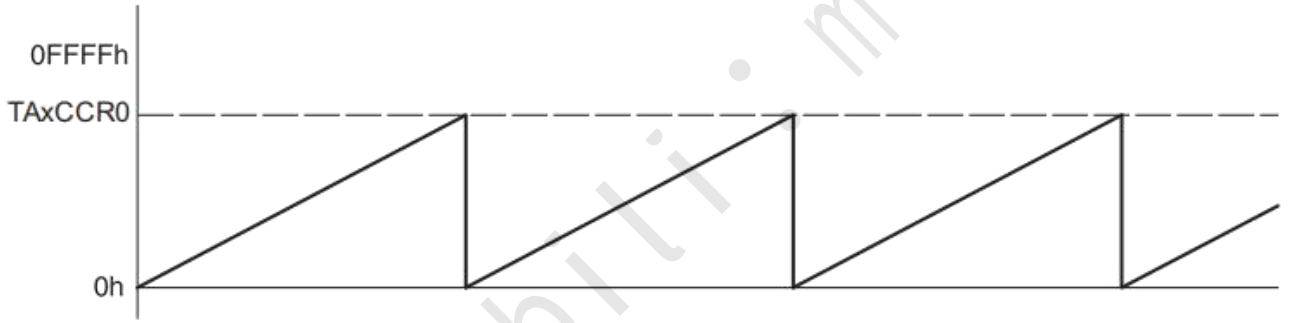


Figure 19-2. Up Mode

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timer_a.h

1 初始化定时器模块

Timer_A_configureUpMode(TIMER_Ax_BASE, &upConfig);

2 选择模式开始计数

Timer_A_startCounter(TIMER_Ax_BASE, TIMER_A_UP_MODE);

3 清除比较中断标志

Timer_A_clearCaptureCompareInterrupt(TIMER_Ax, REGISTER_0);

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interrupt.h

4 开启定时器A端口中断

Interrupt_enableInterrupt(INT_TAx_0);

5 开启总中断

Interrupt_enableMaster(void);

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一般配置步骤

0.配置时钟
1.配置结构体
2.初始化定时器A
3.选择模式开始计数
4.清除比较中断标志位
5.开启定时器端口中断
6.开启总中断
7.编写TIMA ISR

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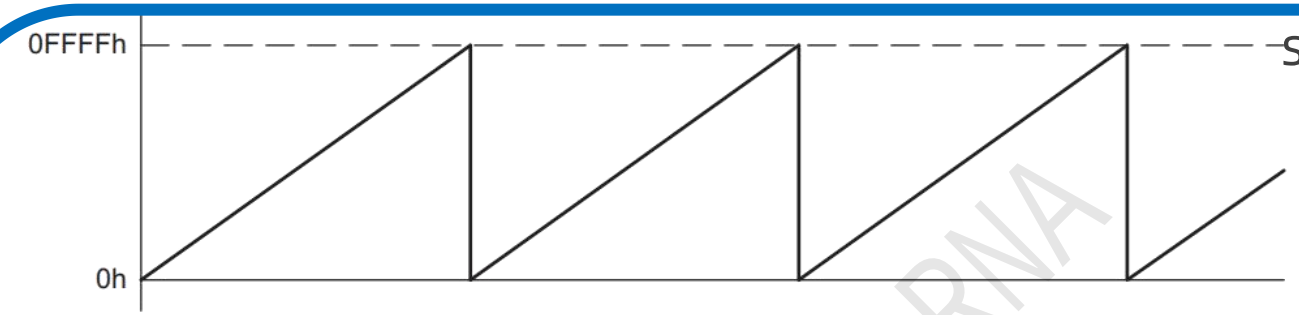


Figure 19-4. Continuous Mode

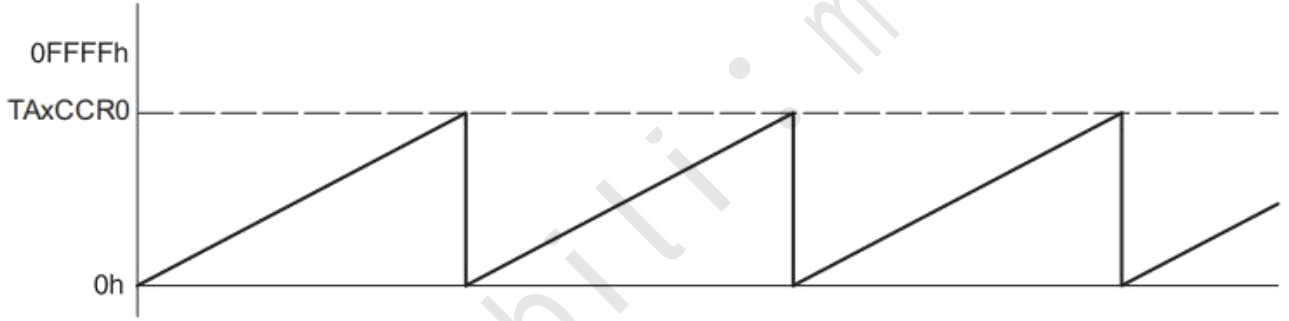


Figure 19-2. Up Mode

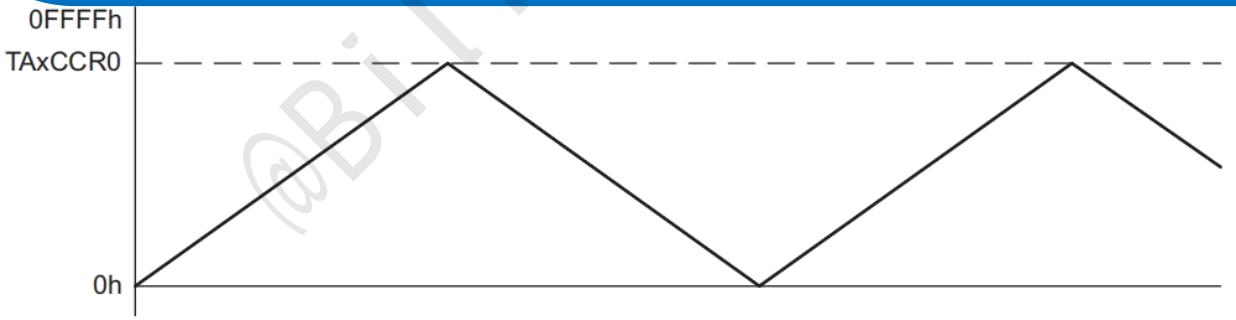


Figure 19-7. Up/Down Mode

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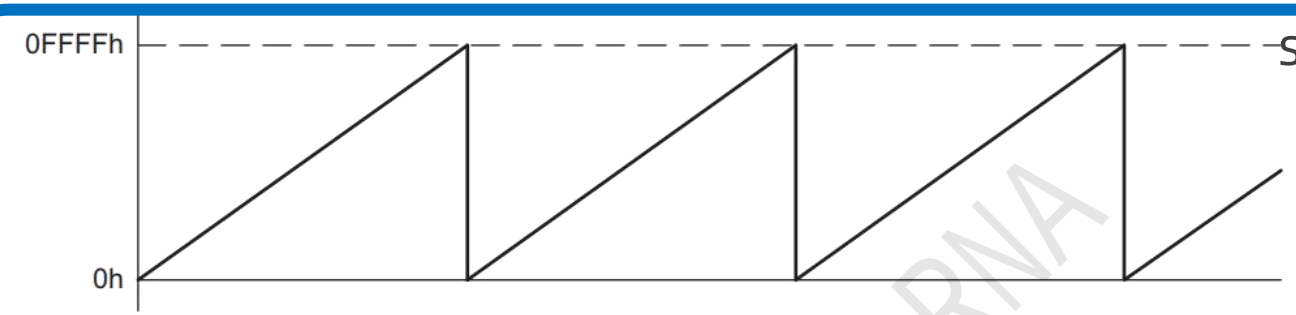


Figure 19-4. Continuous Mode

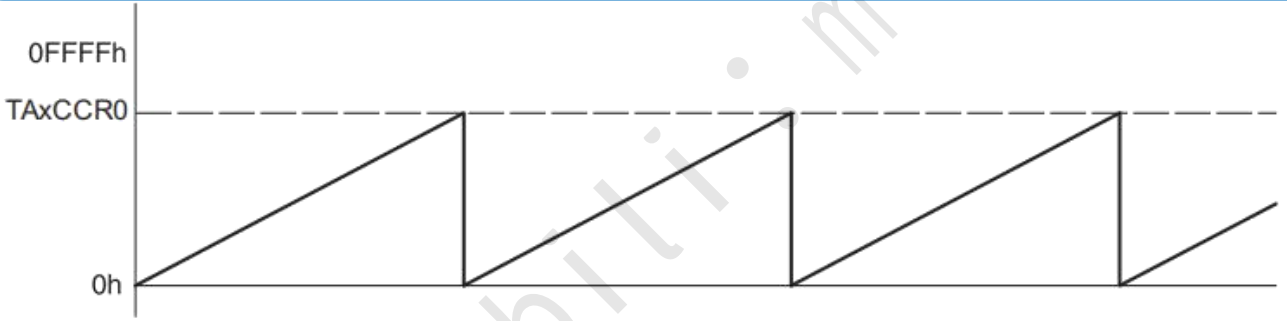


Figure 19-2. Up Mode

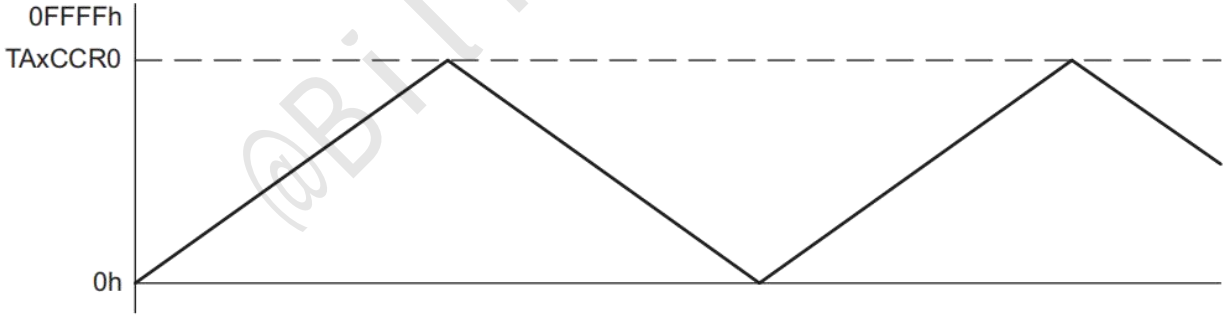


Figure 19-7. Up/Down Mode

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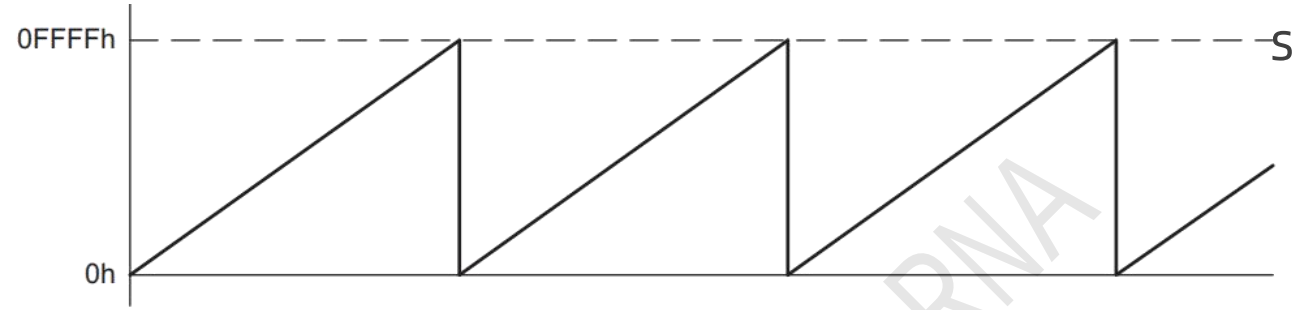


Figure 19-4. Continuous Mode

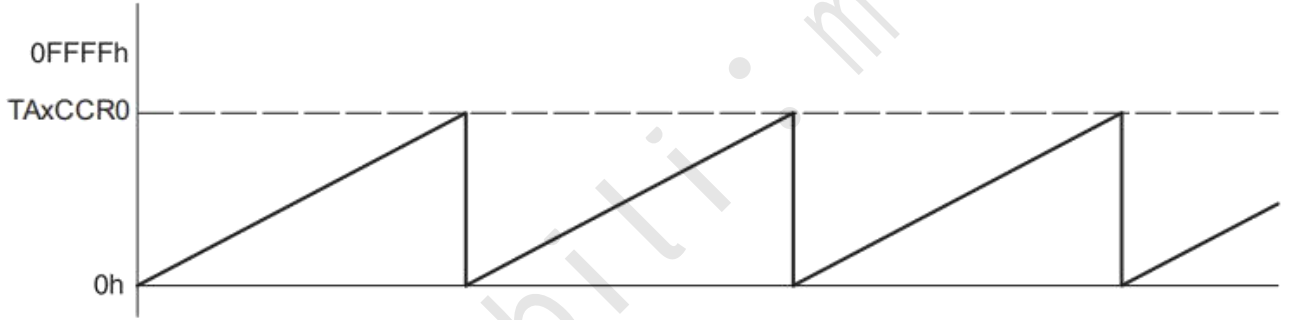


Figure 19-2. Up Mode

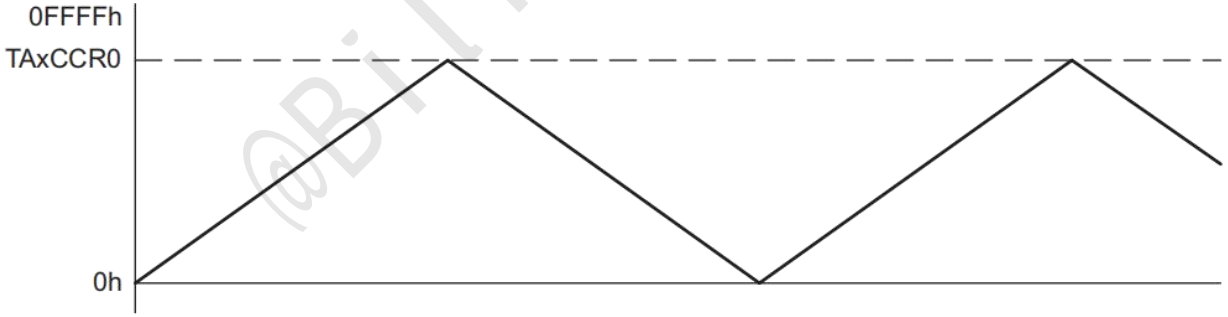


Figure 19-7. Up/Down Mode

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TIMER_A -> INT/PWM

$$T_{timer_a} = \frac{ClkDiv \times (CCR0 + 1)}{f_{Clk}}$$

CLKDIV ∈ [1, 8] U {10, 12, 14, 16, 20, 24, 28, 32, 40, 48, 56, 64 };

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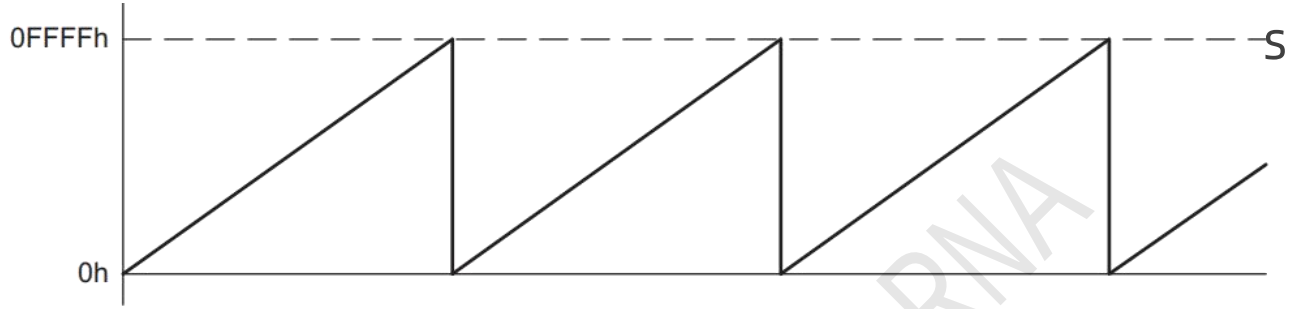


Figure 19-4. Continuous Mode

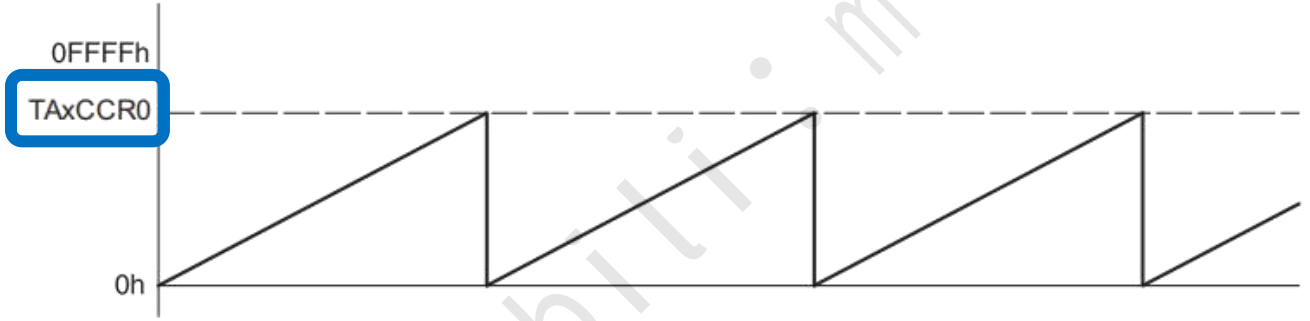


Figure 19-2. Up Mode

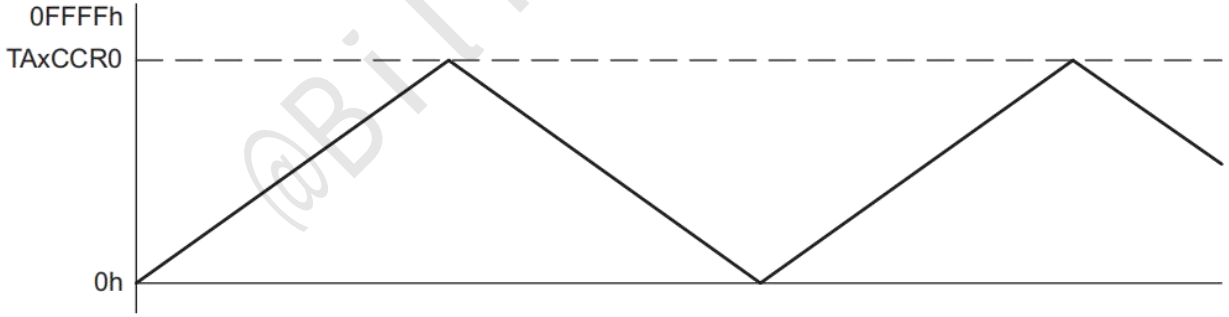


Figure 19-7. Up/Down Mode

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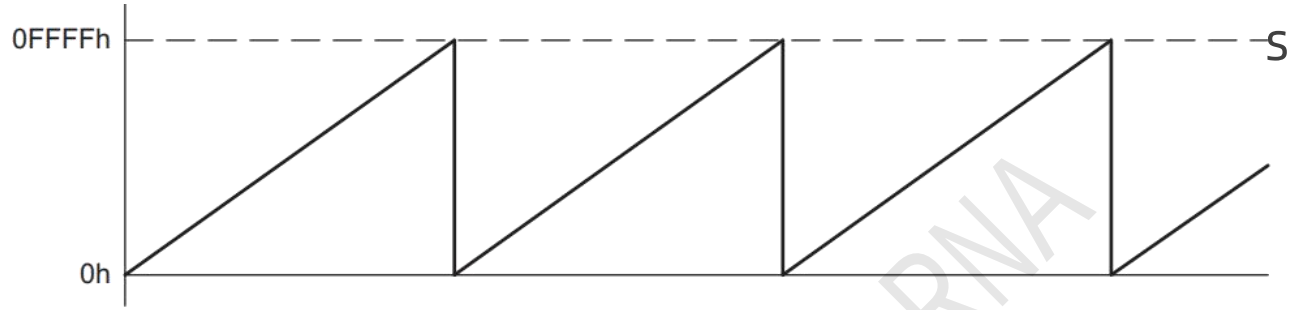


Figure 19-4. Continuous Mode

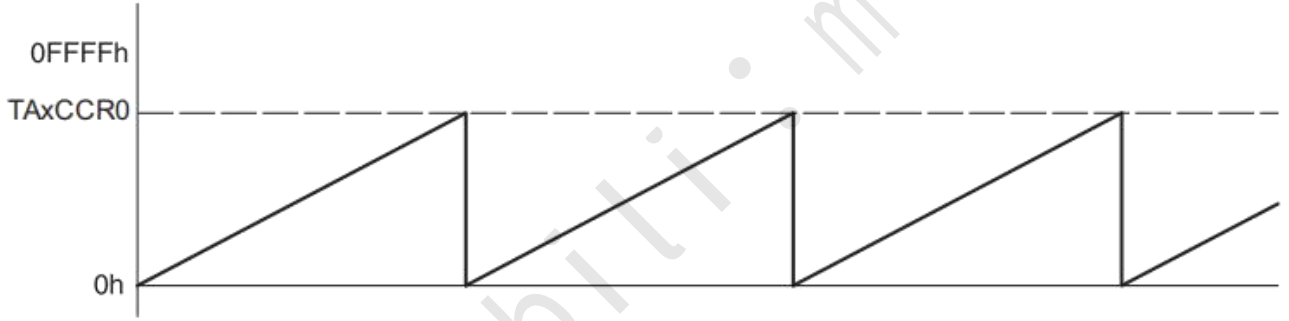


Figure 19-2. Up Mode

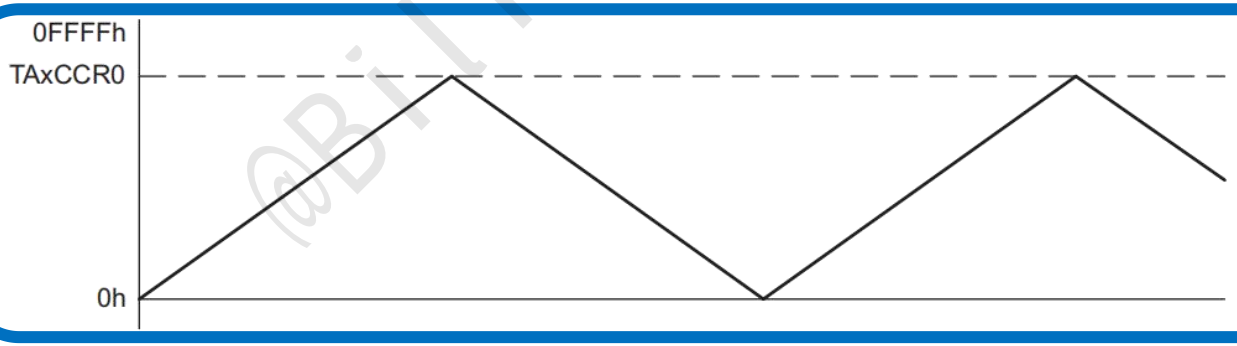


Figure 19-7. Up/Down Mode

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8-2 定时器A PWM

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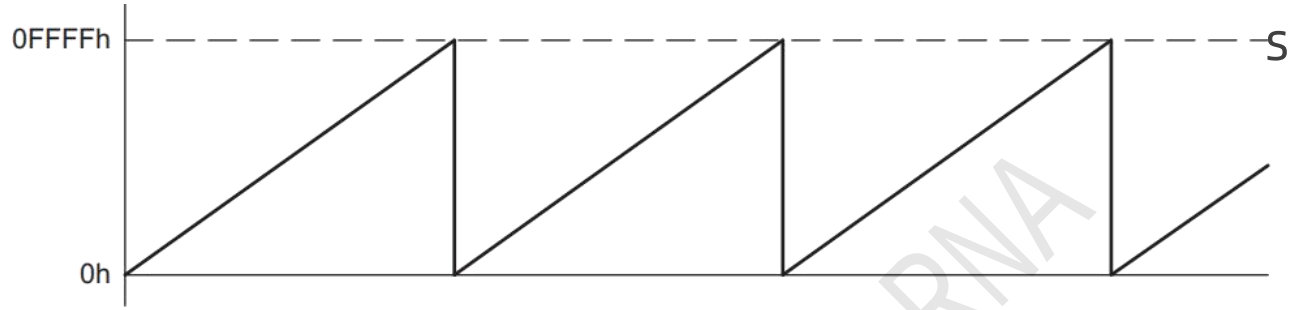


Figure 19-4. Continuous Mode

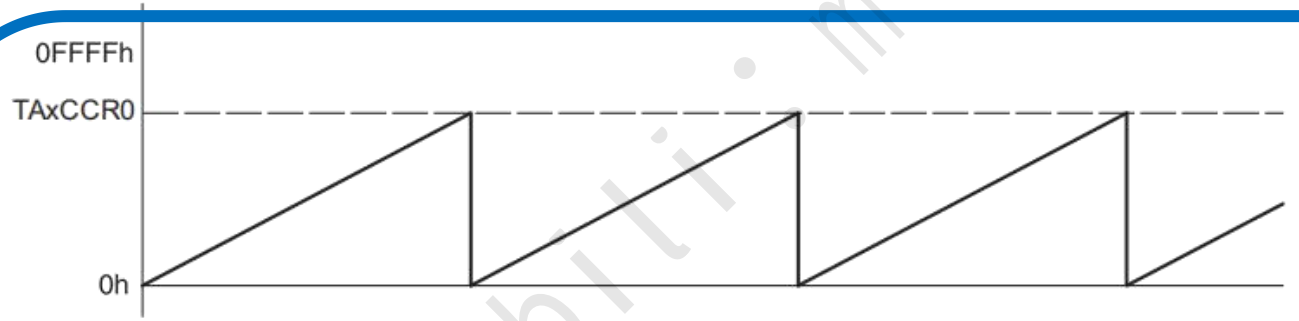


Figure 19-2. Up Mode

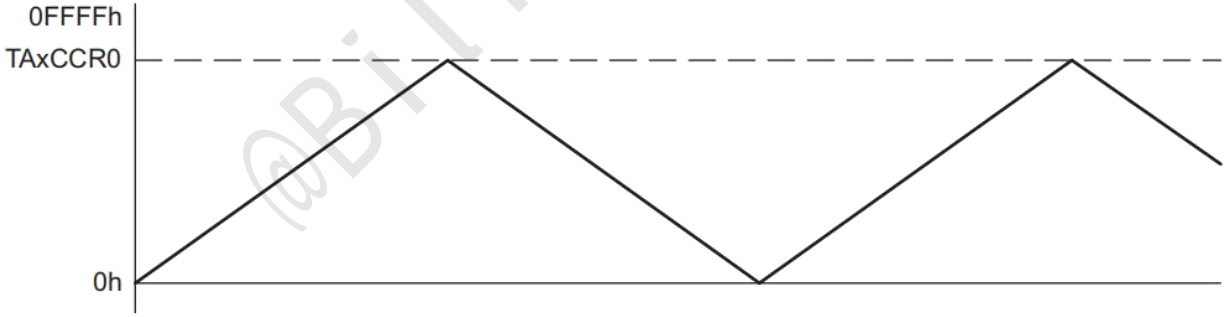


Figure 19-7. Up/Down Mode

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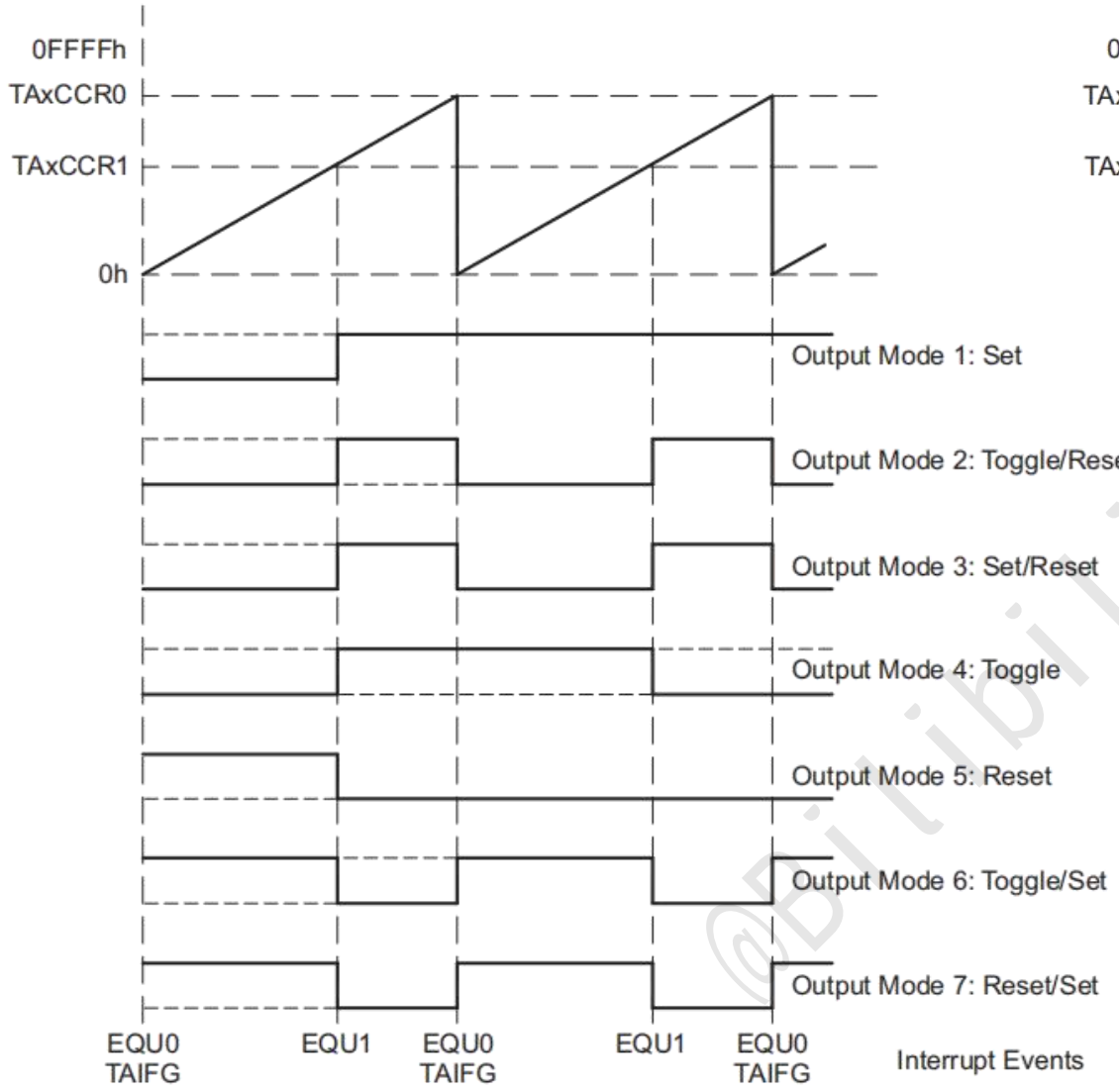


Figure 19-12. Output Example – Timer in Up Mode

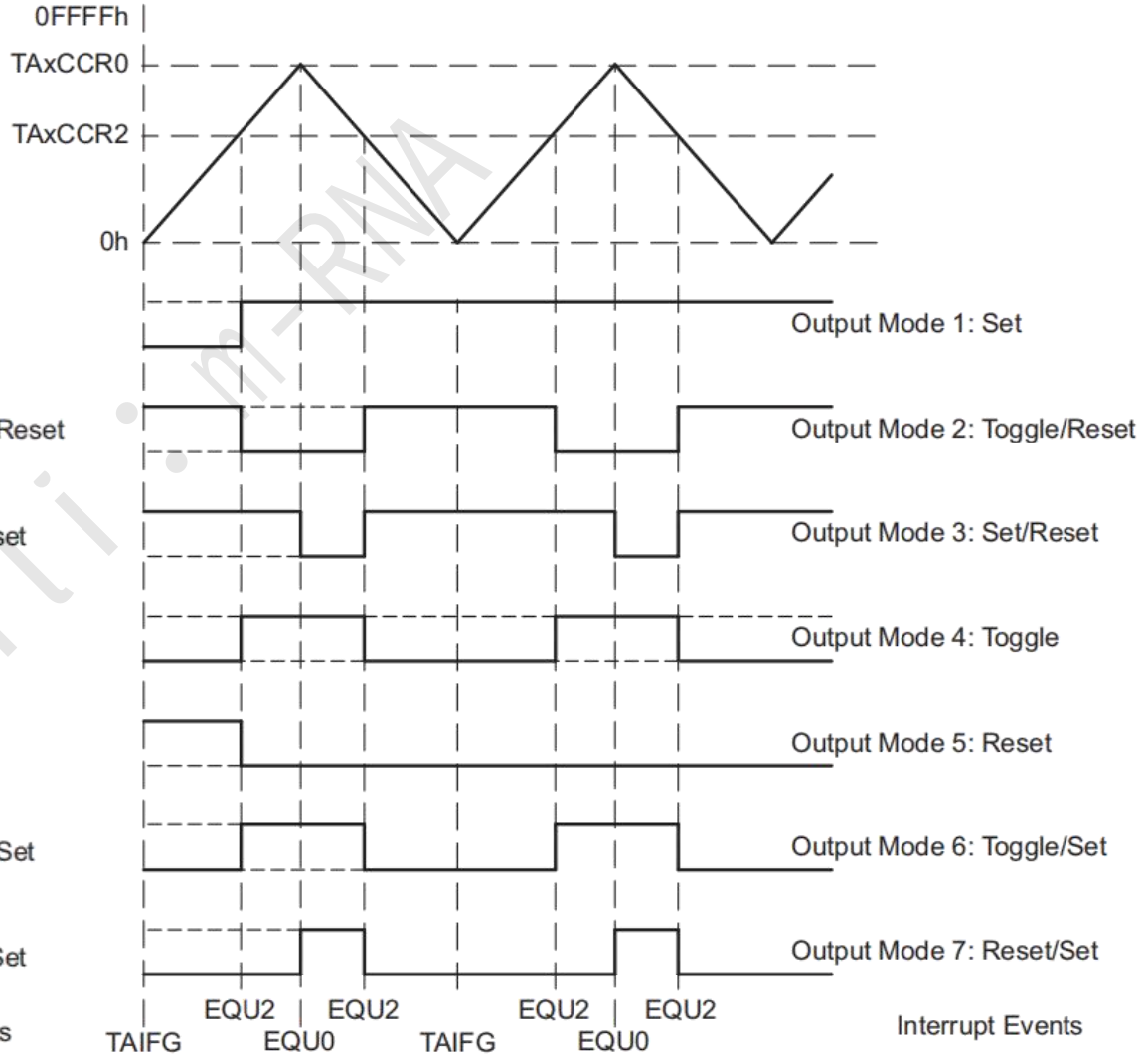
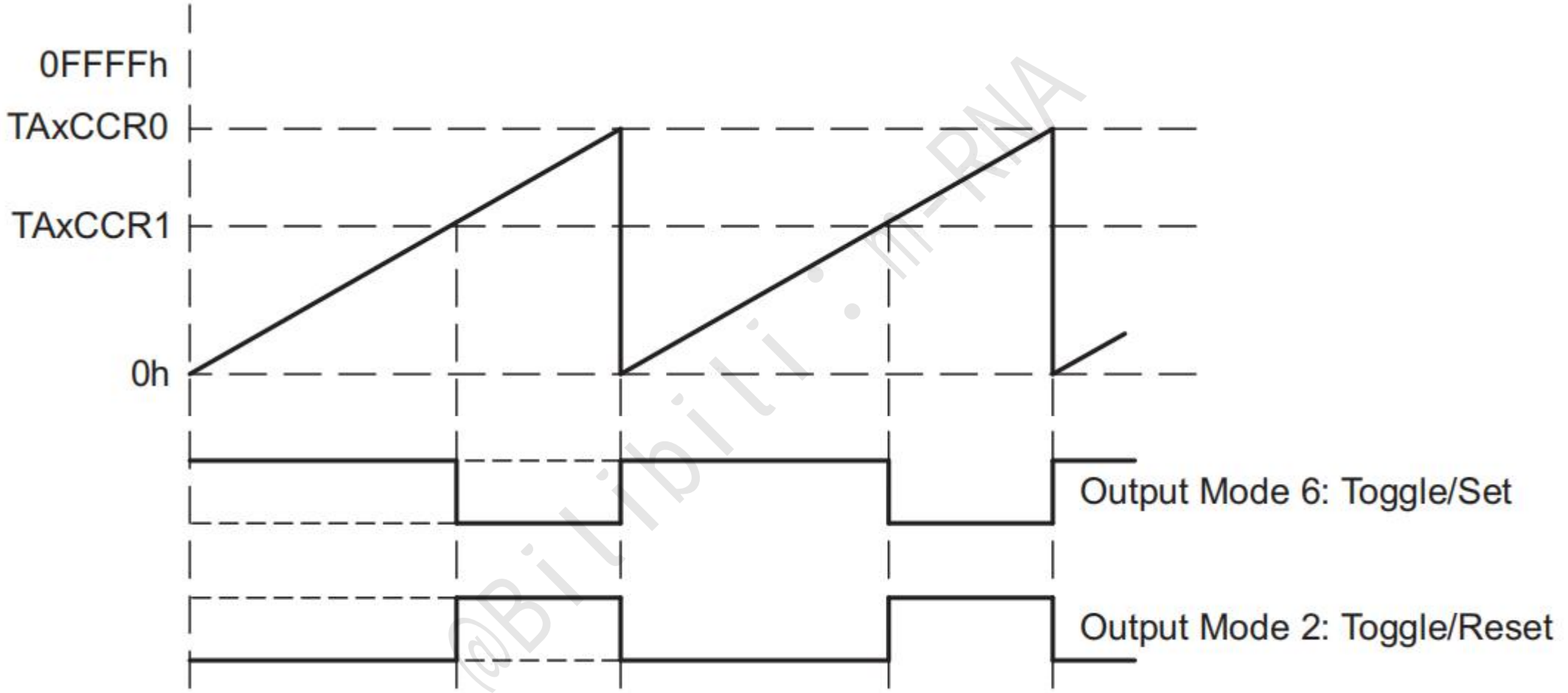
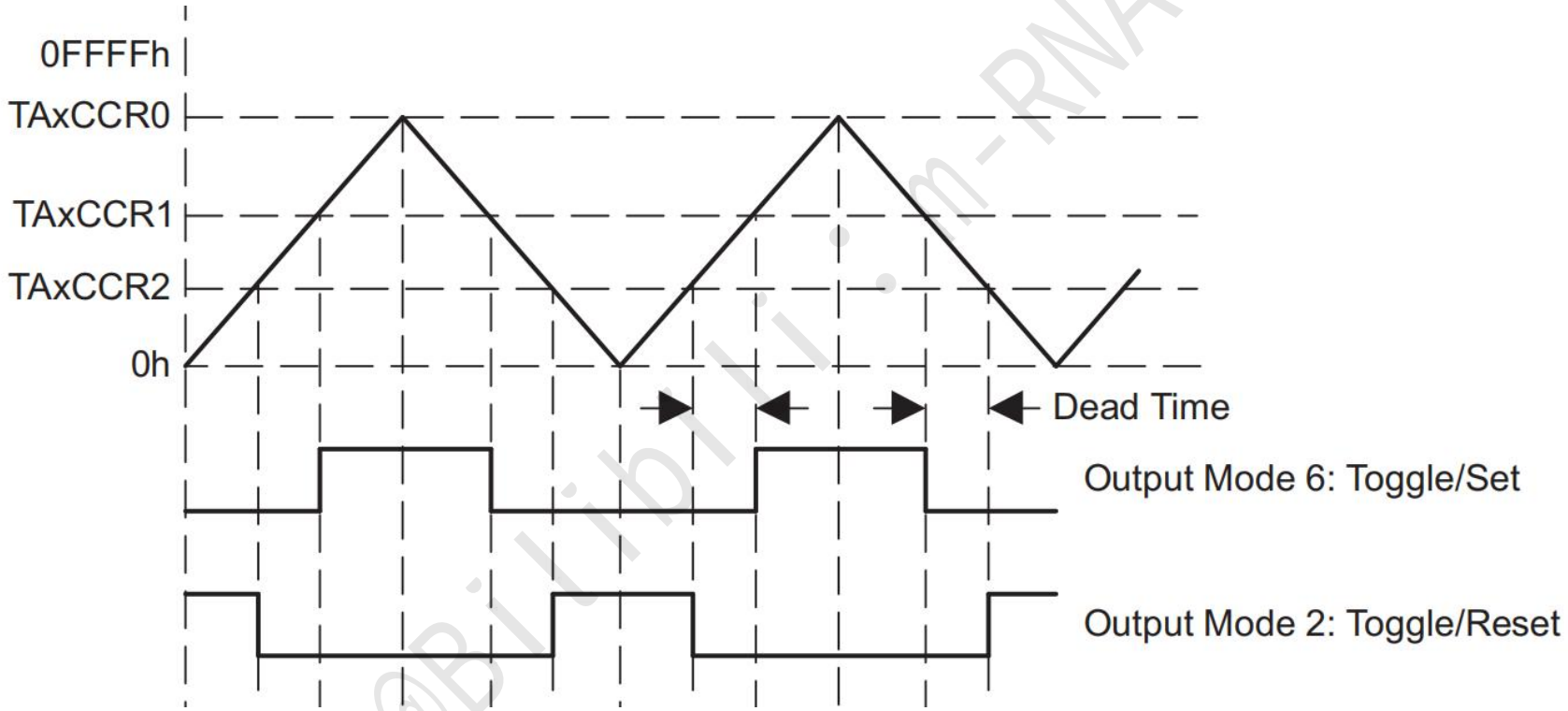
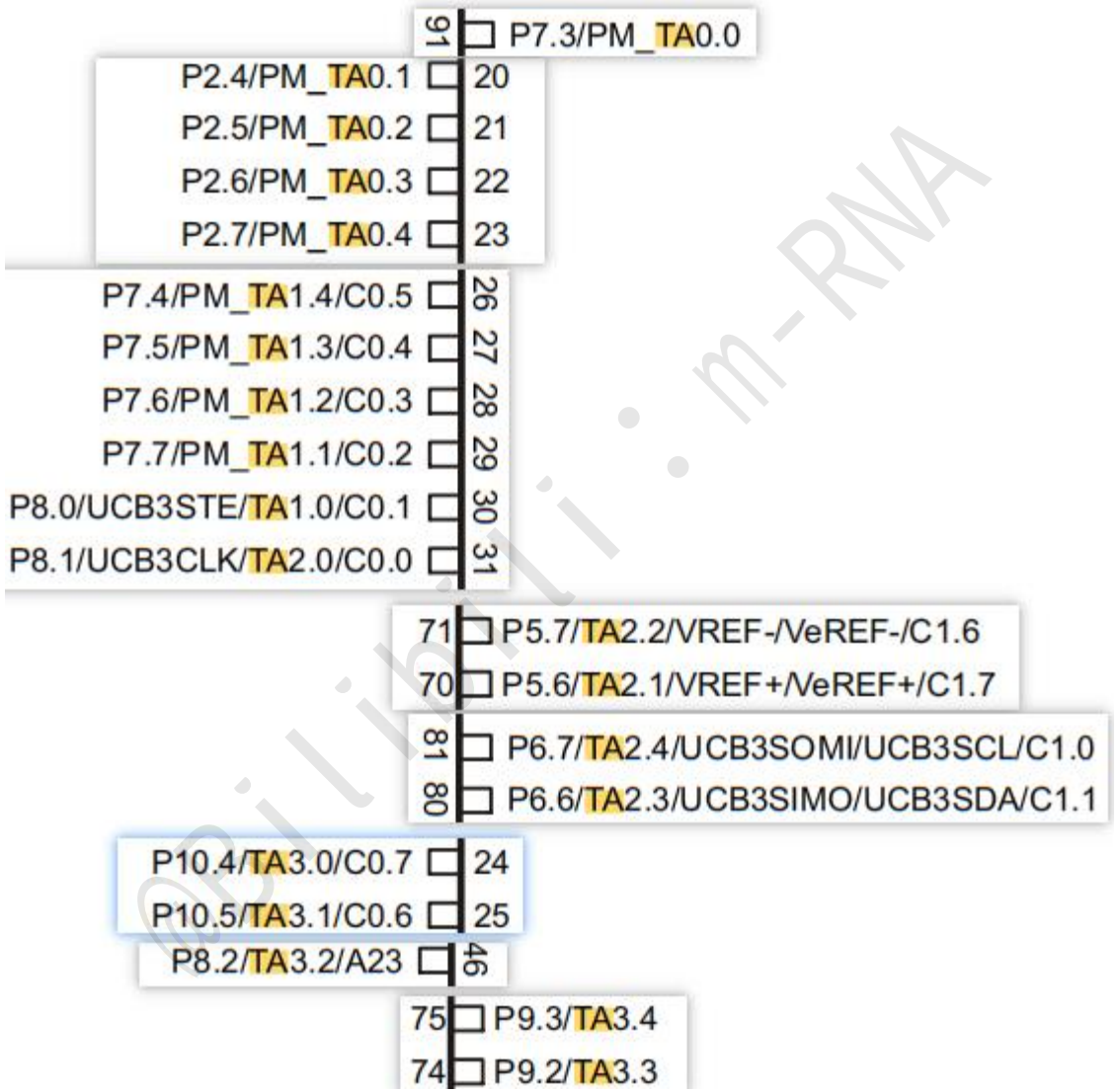


Figure 19-14. Output Example – Timer in Up/Down Mode

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0FFFFh
TAxCCR0
TAxCCR1
0h
Output Mode 6: Toggle/Set
Output Mode 2: Toggle/Reset


0FFFFh
TAxCCR0
TAxCCR1
TAxCCR2
0h
Dead Time
Output Mode 6: Toggle/Set
Output Mode 2: Toggle/Reset


91 P7.3/PM_TA0.0
P2.4/PM_TA0.1 20
P2.5/PM_TA0.2 21
P2.6/PM_TA0.3 22
P2.7/PM_TA0.4 23
P7.4/PM_TA1.4/C0.5 26
P7.5/PM_TA1.3/C0.4 27
P7.6/PM_TA1.2/C0.3 28
P7.7/PM_TA1.1/C0.2 29
P8.0/UCB3STE/TA1.0/C0.1 30
P8.1/UCB3CLK/TA2.0/C0.0 31
71 P5.7/TA2.2/VREF-/VeREF-/C1.6
70 P5.6/TA2.1/VREF+/VeREF+/C1.7
81 P6.7/TA2.4/UCB3SOMI/UCB3SCL/C1.0
80 P6.6/TA2.3/UCB3SIMO/UCB3SDA/C1.1
P10.4/TA3.0/C0.7 24
P10.5/TA3.1/C0.6 25
P8.2/TA3.2/A23 46
75 P9.3/TA3.4
74 P9.2/TA3.3

MSP432P401R 定时器A					
TAx.x	**0**	**1**	**2**	**3**	**4**
TA0	P7.3	P2.4	P2.5	P2.6	P2.7
TA1	P8.0	P7.7	P7.6	P7.5	P7.4
TA2	P8.1	P5.6	P5.7	P6.6	P6.7
TA3	P10.4	P10.5	P8.2	P9.2	P9.3

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timer_a.h

1 初始化定时器为PWM模式

Timer_A_generatePWM(TIMER_Ax_BASE, &TimAx_PWMConfig);

2 改变比较值（占空比/周期）

Timer_A_setCompareValue(TIMER_Ax, COMPARE_REGISTER_x, CCR);

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一般配置步骤

0.配置时钟

1.配置GPIO复用

2.配置结构体

3.初始化定时器

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PIN NAME (P7.x)	x	FUNCTION	CONTROL BITS OR SIGNALS[1]			
			P7DIR.x	P7SEL1.x	P7SEL0.x	P7MAPx
P7.4/PM_TA1.4/C0.5[2]	4	P7.4 (I/O)	I: 0; O: 1	0	0	X
		TA1.CCI4A	0	0	1	default
		TA1.4	1			
		N/A	0	1	0	X
		DVSS	1			
		C0.5[3][4]	X	1	1	X
P7.5/PM_TA1.3/C0.4[2]	5	P7.5 (I/O)	I: 0; O: 1	0	0	X
		TA1.CCI3A	0	0	1	default
		TA1.3	1			
		N/A	0	1	0	X
		DVSS	1			
		C0.4[3][4]	X	1	1	X
P7.6/PM_TA1.2/C0.3[2]	6	P7.6 (I/O)	I: 0; O: 1	0	0	X
		TA1.CCI2A	0	0	1	default
		TA1.2	1			
		N/A	0	1	0	X
		DVSS	1			
		C0.3[3][4]	X	1	1	X
P7.7/PM_TA1.1/C0.2[2]	7	P7.7 (I/O)	I: 0; O: 1	0	0	X
		TA1.CCI1A	0	0	1	default
		TA1.1	1			
		N/A	0	1	0	X
		DVSS	1			
		C0.2[3][4]	X	1	1	X

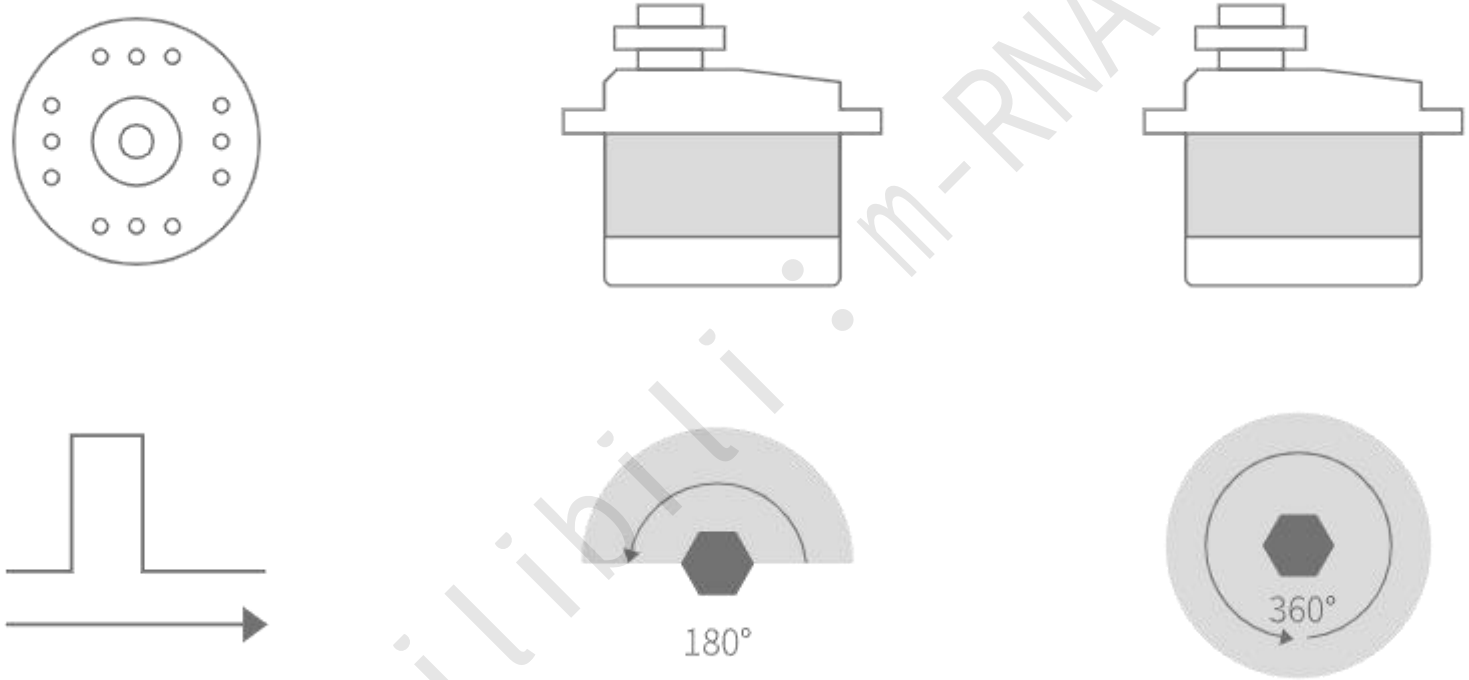
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TIMER_A

$$T_{timer_a} = \frac{ClkDiv \times (CCR0 + 1)}{f_{Clk}}$$

CLKDIV ∈ [1, 8] U {10, 12, 14, 16, 20, 24, 28, 32, 40, 48, 56, 64 };

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频率：50Hz　　占空比：2.5% ~12.5%

8-3 定时器 A 捕获

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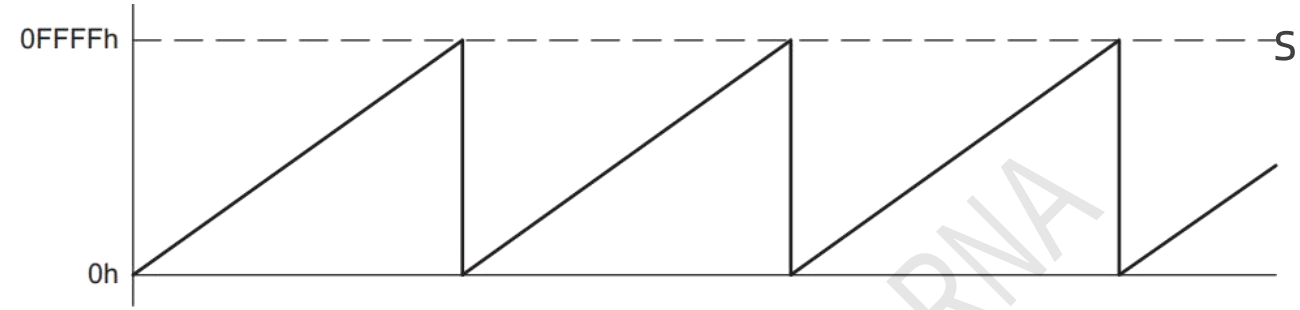


Figure 19-4. Continuous Mode

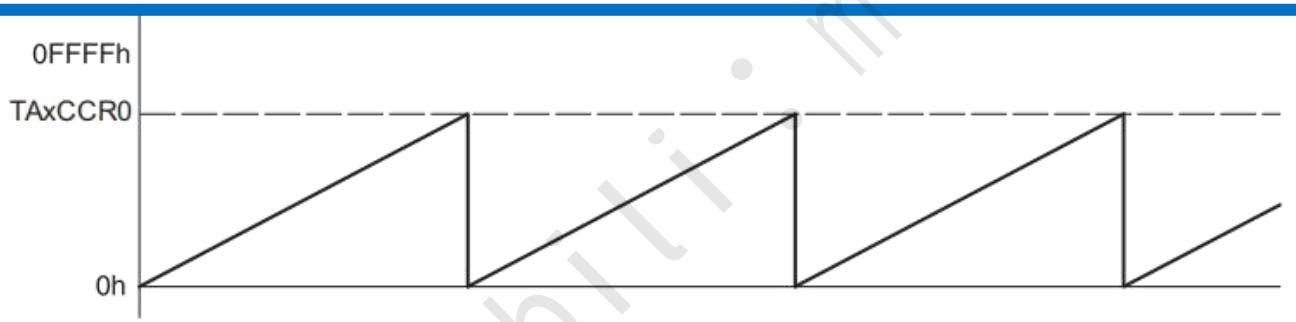


Figure 19-2. Up Mode

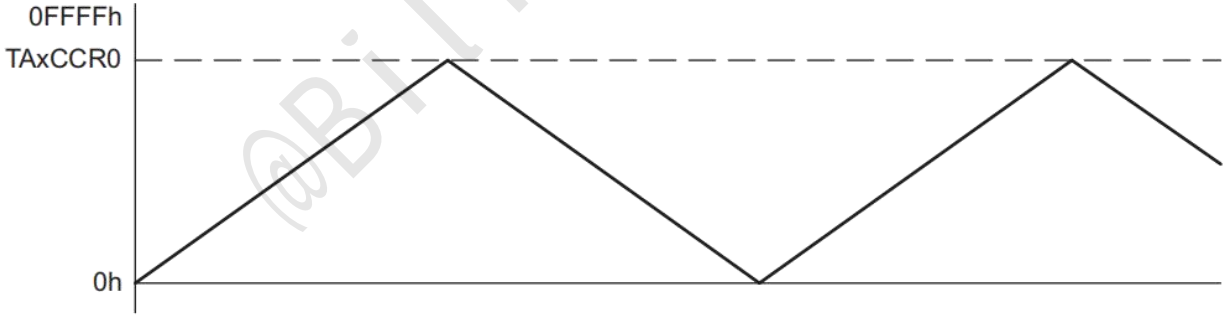


Figure 19-7. Up/Down Mode

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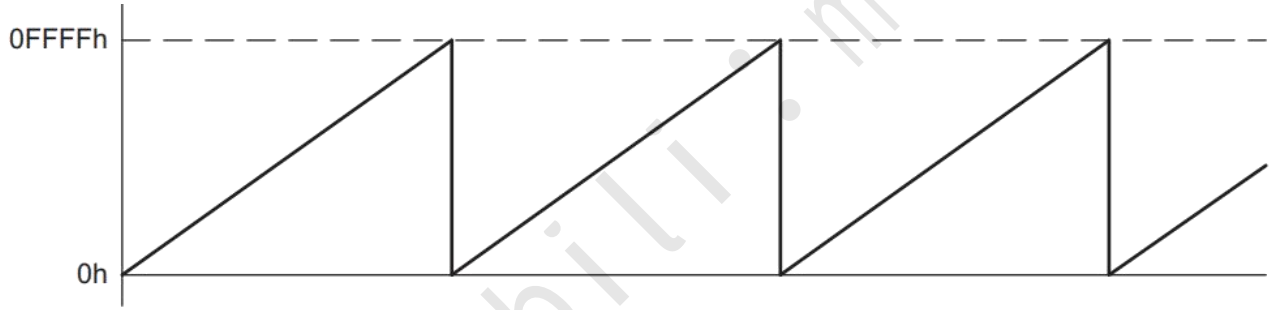


Figure 19-4. Continuous Mode

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timer_a.h

1 初始化定时器为连续计数模式

Timer_A_configureContinuousMode(TIMER_Ax, &continuousModeConfig);

2 配置定时器的捕获模式

Timer_A_initCapture(TIMER_Ax_BASE, &captureModeConfig);

3 选择模式开始计数

Timer_A_startCounter(TIMER_Ax_BASE, TIMER_A_CONTINUOUS_MODE);

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timer_a.h

4 清除定时器溢出中断标志位

Timer_A_clearInterruptFlag(TIMER_Ax_BASE);

5 清除定时器捕获中断标志位

Timer_A_clearCaptureCompareInterrupt(TIMER_Ax, REGISTER_N);

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timer_a.h

6 获取定时器溢出中断状态

Timer_A_getEnabledInterruptStatus(TIMER_Ax_BASE);

7 获取定时器捕获中断状态

Timer_A_getCaptureCompareEnabledInterruptStatus(TIMER_Ax, REGISTER_N);

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timer_a.h

8 获取定时器捕获电平状态

Timer_A_getSynchronizedCaptureCompareInput(TIMER_Ax, REGISTER_N, Setting);

返回值:

TIMER_A_OUTPUTMODE_OUTBITVALUE_LOW
TIMER_A_OUTPUTMODE_OUTBITVALUE_HIGH

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interrupt.h

9 开启定时器A端口中断

Interrupt_enableInterrupt(INT_TAx_N);

10 开启总中断

Interrupt_enableMaster(void);

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一般配置步骤

0.配置时钟
1.复用引脚
2.配置连续计数结构体
3.初始化定时器连续计数
4.配置捕获结构体
5.初始化定时器为捕获
6.选择模式开始计数
7.清除中断标志位
8.开启定时器端口中断
9.开启总中断
10.编写TIMA ISR

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TIMER_A -> CAP

$$t_{cap} = \frac{ClkDiv \times (CCRN + 1)}{f_{clk}}$$

CLKDIV ∈ [1, 8] U {10, 12, 14, 16, 20, 24, 28, 32, 40, 48, 56, 64 };

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表 6-75. Port P5 (P5.6 and P5.7) Pin Functions

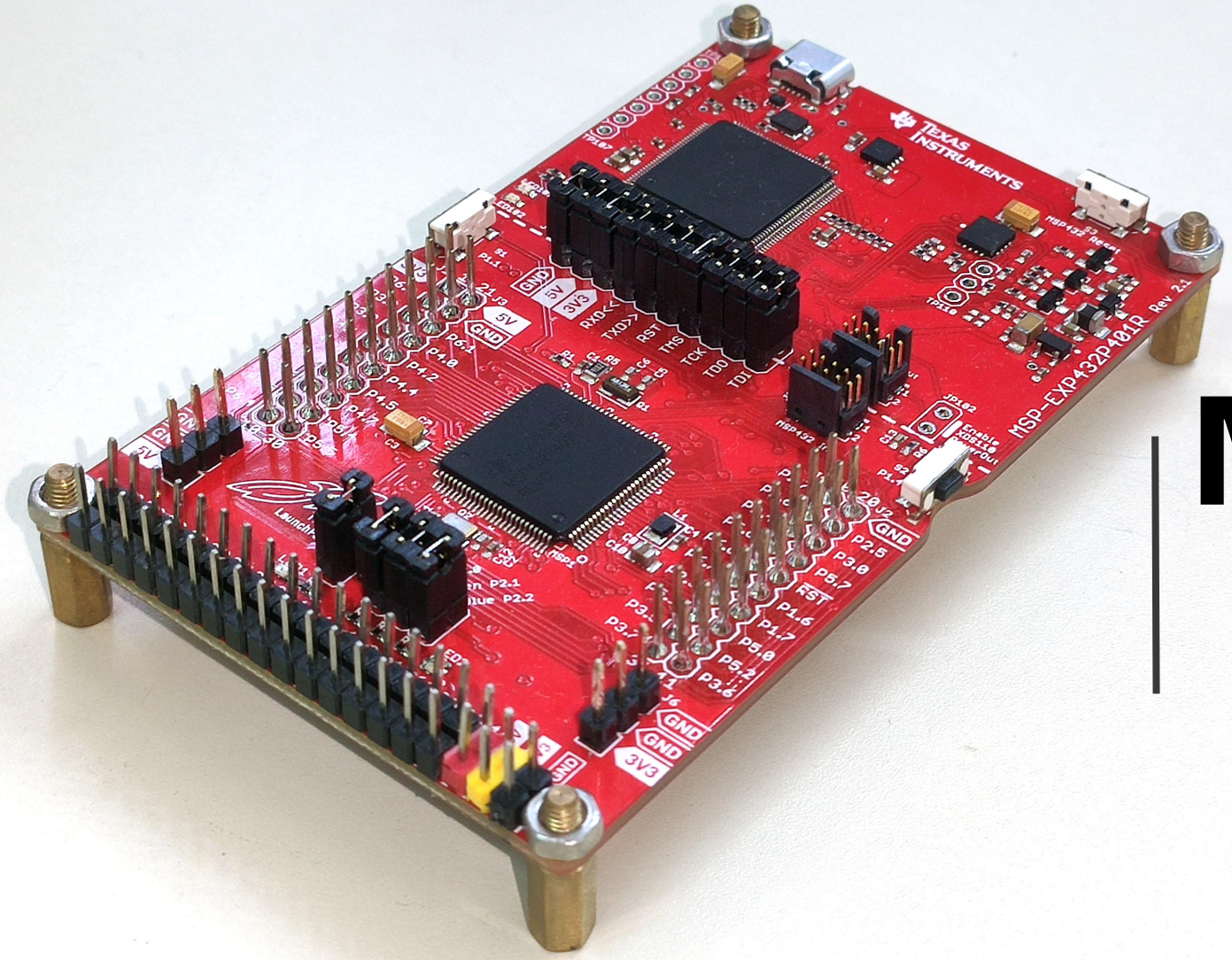
PIN NAME (P5.x)	x	FUNCTION	CONTROL BITS OR SIGNALS(1)		
			P5DIR.x	P5SEL1.x	P5SEL0.x
P5.6/TA2.1/VREF+/VeREF+/C1.7	6	P5.6 (I/O)	I: 0; O: 1	0	0
		TA2.CCI1A	0	0	1
		TA2.1	1		
		N/A	0	1	0
		DVSS	1		

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2021/10

第9讲 定时器32

参考资料

slau356h.pdf　　P756

18.1 Introduction

The Timer32 is an AMBA compliant peripheral developed, tested, and licensed by ARM Limited. The Timer32 consists of two programmable 32-bit or 16-bit down counters that can generate interrupts on reaching zero.

The key features of Timer32 include:

- Two independent counters each configurable as 32-bit or 16-bit counter size
- Three different timer modes supported for each counter
- Prescale unit to divide the input clock by 1, 16 or 256
- Independent Interrupts from each of the counter, as well as, a combined interrupt from both the counters

Timer32的主要特性包括:

◇两个独立的计数器,每个都可配置成32位或16位计数器;

◇每个计数器具有3种不同的定时器模式;

◇每个计数器都可独立产生中断,而且两个计数器可生成一个组合中断。

◇输入时钟可预分频为1、1/16或1/256;(MCLK)

中断向量:

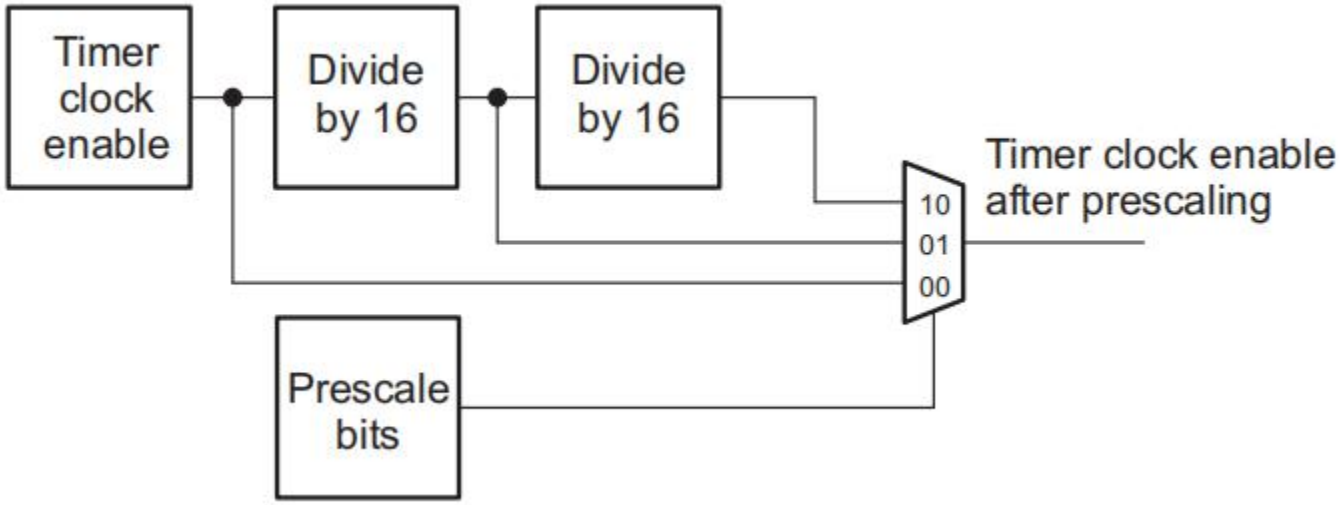
◇INT_T32_INT1（定时器32_0）

◇INT_T32_INT2（定时器32_1）

◇INT_T32_INTC（Combine 结合)

The timer clock enable is generated by a prescale unit. The enable is then used by the counter to create a clock with a timing of one of the following:

- MCLK
- MCLK divided by 16, generated by 4 bits of prescale
- MCLK divided by 256, generated by a total of 8 bits of prescale

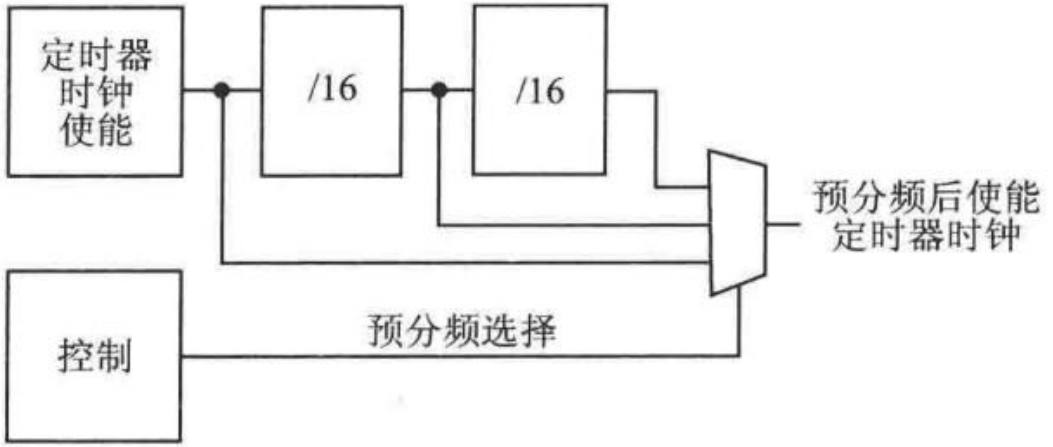


定时器时钟使能由分频单元产生,并使能由计数器创建的具有下列条件之一的定时时钟:

◇MCLK #define TIMER32_PRESCALER_1 0x00

◇由4位预分频产生的16分频MCLK #define TIMER32_PRESCALER_16 0x04

◇由总共8位预分频产生的256分频MCLK #define TIMER32_PRESCALER_256 0x08



18.2 Functional Description

Two independent timers are available in Timer32. For each of the timers, the following modes of operation are available:

- Free-running mode: The counter wraps after reaching its zero value, and continues to count down from the maximum value. This is the default mode.
- Periodic timer mode: The counter generates an interrupt at a constant interval, reloading the original value after wrapping past zero.
- One-shot timer mode: The counter generates an interrupt once. When the counter reaches zero, it halts until reprogrammed by the user. This can be achieved by either clearing the One Shot Count bit in the control register, in which case the count proceeds according to the selection of free-running or periodic mode, or by writing a new value to the Load Value register.

Timer32的计数模式:

◇自由运行模式： 计数器在递减到0后，继续从最大值递减。这是默认模式。

◇周期定时器模式：计数器以恒定的间隔生成一个中断，在递减到0后重新加载原始值。

◇单次定时器模式：计数器产生一次中断。当计数器达到零时，它会停止，直到被用户重新编程。

TIMER_32 -> INT

$$T_{timer_32} = \frac{ClkDiv \times (ARR + 1)}{f_{MClk}}$$

ClkDiv ∈ {1, 16, 256 };

一般配置步骤

0.配置时钟
1.初始化为32位周期计数模式
2.设置ARR自动重装载值
3.清除中断标志位
4.使能定时器32中断
5.配置定时器32开始连续计数
6.开启定时器32端口中断
7.开启总中断
8.编写TIM32 ISR



参 考 书 籍

基于固件的MSP432微控制器
原理及应用
刘杰 著

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